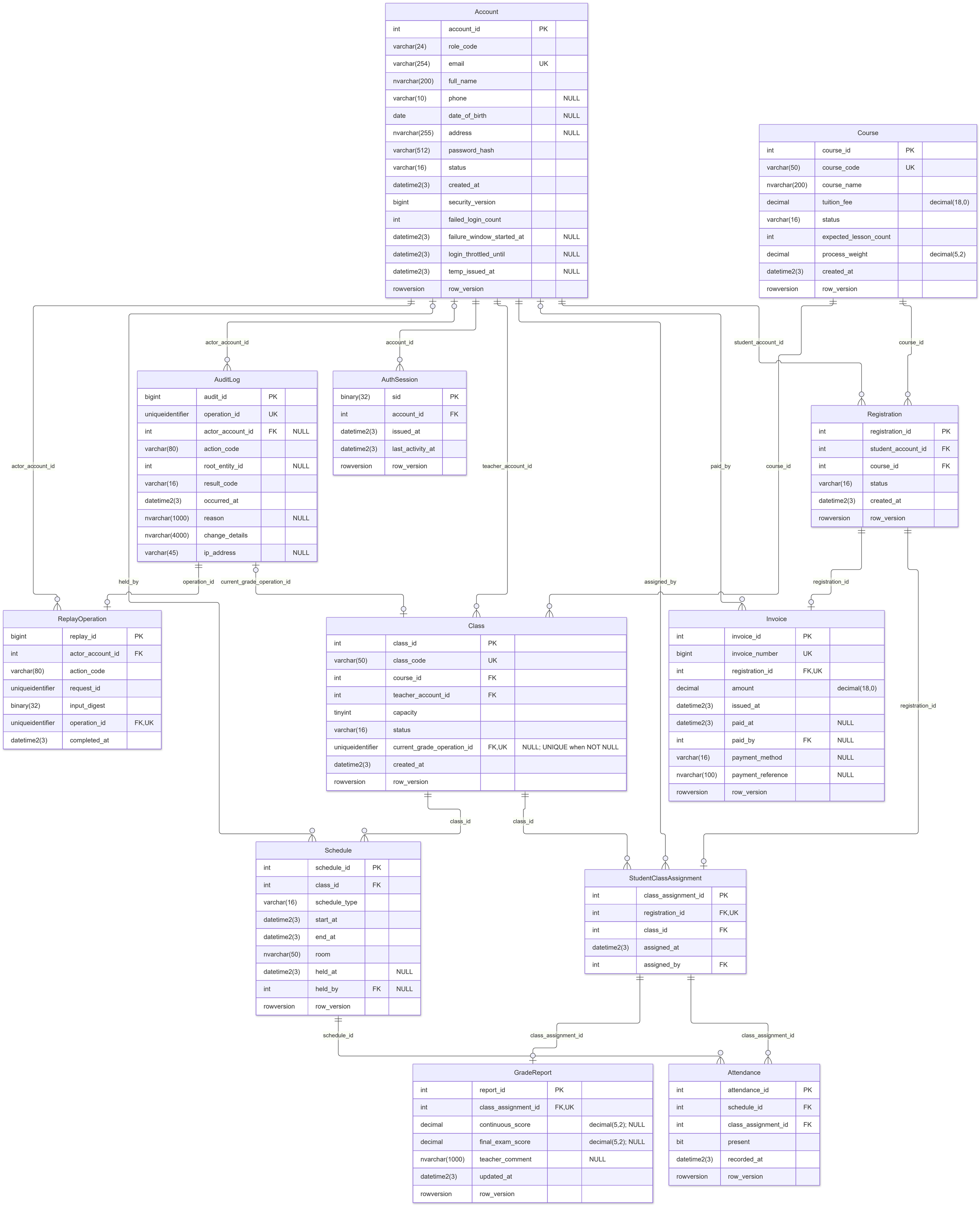

Account
int account_id PK
varchar(24) role_code
varchar(254) email UK
nvarchar(200) full_name
varchar(10) phone NULL
date date_of_birth NULL
nvarchar(255) address NULL
varchar(512) password_hash
varchar(16) status
datetime2(3) created_at
bigint security_version
int failed_login_count
datetime2(3) failure_window_started_at NULL
datetime2(3) login_throttled_until NULL
datetime2(3) temp_issued_at NULL
rowversion row_version
Course
int course_id PK
varchar(50) course_code UK
nvarchar(200) course_name
decimal tuition_fee decimal(18,0)
varchar(16) status
int expected_lesson_count
decimal process_weight decimal(5,2)
datetime2(3) created_at
rowversion row_version
actor_account_id
account_id
student_account_id
course_id
AuditLog
bigint audit_id PK
uniqueidentifier operation_id UK
int actor_account_id FK NULL
varchar(80) action_code
int root_entity_id NULL
varchar(16) result_code
datetime2(3) occurred_at
nvarchar(1000) reason NULL
nvarchar(4000) change_details
varchar(45) ip_address NULL
AuthSession
binary(32) sid PK
int account_id FK
datetime2(3) issued_at
datetime2(3) last_activity_at
rowversion row_version
Registration
int registration_id PK
int student_account_id FK
int course_id FK
varchar(16) status
datetime2(3) created_at
rowversion row_version
actor_account_id
teacher_account_id
paid_by
course_id
held_by
operation_id
current_grade_operation_id
assigned_by
registration_id
ReplayOperation
bigint replay_id PK
int actor_account_id FK
varchar(80) action_code
uniqueidentifier request_id
binary(32) input_digest
uniqueidentifier operation_id FK,UK
datetime2(3) completed_at
Class
int class_id PK
varchar(50) class_code UK
int course_id FK
int teacher_account_id FK
tinyint capacity
varchar(16) status
uniqueidentifier current_grade_operation_id FK,UK NULL; UNIQUE when NOT NULL
datetime2(3) created_at
rowversion row_version
Invoice
int invoice_id PK
bigint invoice_number UK
int registration_id FK,UK
decimal amount decimal(18,0)
datetime2(3) issued_at
datetime2(3) paid_at NULL
int paid_by FK NULL
varchar(16) payment_method NULL
nvarchar(100) payment_reference NULL
rowversion row_version
registration_id
class_id
class_id
Schedule
int schedule_id PK
int class_id FK
varchar(16) schedule_type
datetime2(3) start_at
datetime2(3) end_at
nvarchar(50) room
datetime2(3) held_at NULL
int held_by FK NULL
rowversion row_version
StudentClassAssignment
int class_assignment_id PK
int registration_id FK,UK
int class_id FK
datetime2(3) assigned_at
int assigned_by FK
schedule_id
class_assignment_id
class_assignment_id
GradeReport
int report_id PK
int class_assignment_id FK,UK
decimal continuous_score decimal(5,2); NULL
decimal final_exam_score decimal(5,2); NULL
nvarchar(1000) teacher_comment NULL
datetime2(3) updated_at
rowversion row_version
Attendance
int attendance_id PK
int schedule_id FK
int class_assignment_id FK
bit present
datetime2(3) recorded_at
rowversion row_version